SUBJECT: Metallurgical Laboratory Report No. W4

TITLE: Examination of Two Worn Ni-Hard Plates Returned by Park 500 of Chester, Virginia

DATE INITIATED: 1/30/92 **DATE COMPLETED:** 2/5/92

Introduction:

1. Two ni-hard refiner plates, pattern D99E901, were returned by Park 500, Chester, Virginia for examination. The plates had been severely worn, and the customer had experienced a substantial drop in plate life. It was reported that consulting metallurgists had measured hardness values of Rc44/46 on similar plates. Those hardnesses are well below the required values for ni-hard, and Park 500 perceived this as the cause of low plate life.

Object:

2. Determine conformance to requirements.

Conclusions:

3. The plates conformed to required metallurgical properties including hardness.

4. The plate surfaces were etched indicating some chemical attack.

5. The low plate life does appear to be the result of low plate hardness.

Results and Discussion:

6. A visual examination of the two refiner plates revealed that they had been worn to a remaining tooth depth of approximately 1/16 inch (Photo 3372). The plate surfaces including the bar faces had a silvery etched appearance. This contrasted sharply with the normal dull gray of as-cast surfaces and normal smooth reflective wear surfaces. Some surface scratching typical of prior metal to metal contact was observed on the bar surfaces, but there was no evidence of severe plastic flow of these surfaces. The leading edge of the bars was rounded and there was a slight taper of the bar surface towards the leading edge. Some residual gas holes were evident.

7. A macroscopic examination of the plates emphasized the effects of chemical etching in service. The matrix had been lightly attacked providing relief to the dendrites that developed during initial crystal formation during solidification. There were no cracks in the bars indicating that suspected prior metal to metal contact had been mild.

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8. The chemistry of the two plates conformed to Andritz Sprout-Bauer Standard 912.07, as illustrated by Table 1.

Table 1. Chemistry of Worn Refiner Plates

	%C	%Mn	%Si	%Cr	%Ni	%Mo	%P	%S	%Cu	%Al	%V
Plate 1	3.43	0.32	0.66	2.45	4.21	0.14	0.024	0.011	0.09	0.006	0.03
Plate 2	3.40	0.32	0.65	2.48	4.23	0.14	0.023	0.011	0.09	0.006	0.03
Std. 912.07	3.00/ 3.65	<0.60	<0.80	2.20/ 3.00	3.70/ 4.70	<0.50	<0.10	<0.05	<0.30	---	<0.05

The manufacturing code for both plates was X302 indicating that they had been cast on the same day. The analyses are so familiar that they were very likely cast from the same heat.

9. At least 10 random hardness readings were taken on bar surfaces of each plate. Care was exercised to select sites that did have excessive tooth taper. In addition, anvils were chosen to minimize any effects of non-parallel surfaces. These simple basic rules, which are well established for hardness testing, are very important for high Rc values because of the shallow indentations produced. The results of these hardness surveys are reported in Figure 1. All of the readings exceeded the specification minimum of Rc55. The values of Rc44/46 reported by the customer are highly irregular for the grade of ni-hard produced by Andritz Sprout-Bauer and the thickness of these refiner plates. This chemistry is designed to develop a hardness of Rc55 in a three-inch thick section.

10. A microexamination of the two plates revealed a structure typical of Andritz Sprout-Bauer ni-hard (Photos 3373 and 3374). There was no evidence of accelerated matrix corrosion, which suggests that the wear rate does not vary significantly from the rate of etching. The microstructures are typical of good ni-hard. The sections were selected from plate bars adjacent to the wear surface.

11. The results of this examination have indicated that the plate specimens conformed to established metallurgical requirements for ni-hard refiner plates. There was no indication of low hardness on either plate. The observed mild corrosive attack may have accelerated the wear rate of the plates.

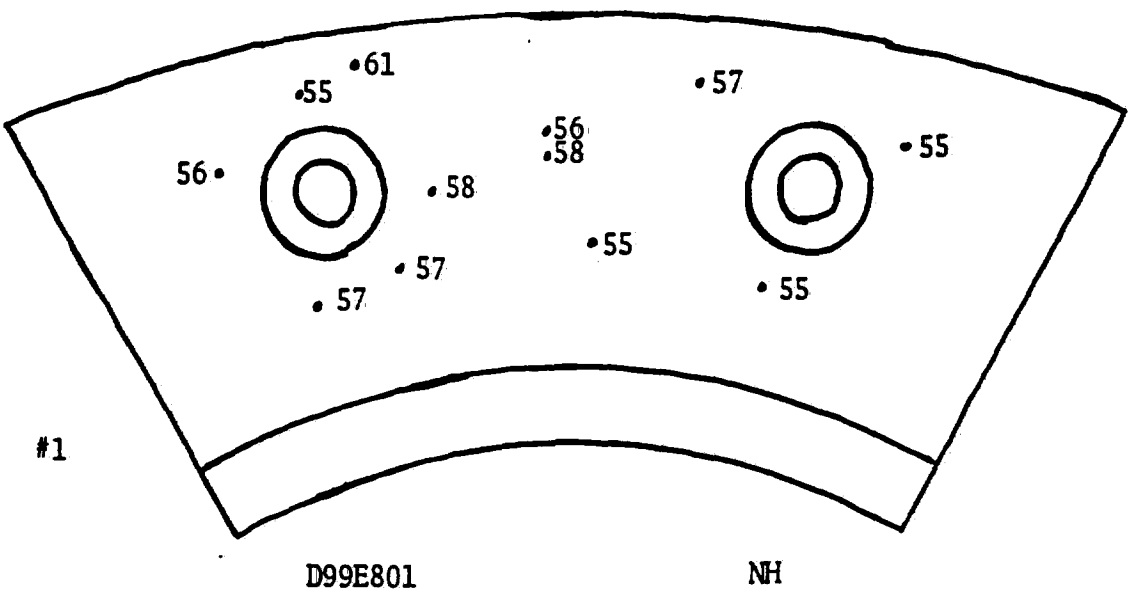
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Refiner Plates

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PROJECT W4



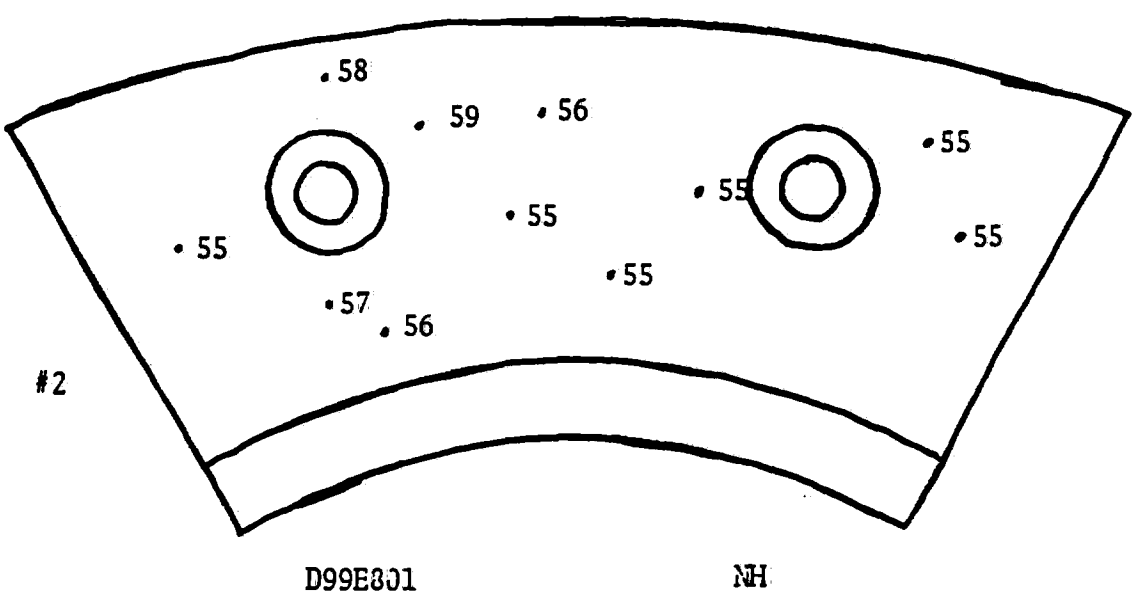


Figure 1. Hardness Survey of Refiner Plates Returned by Park 500

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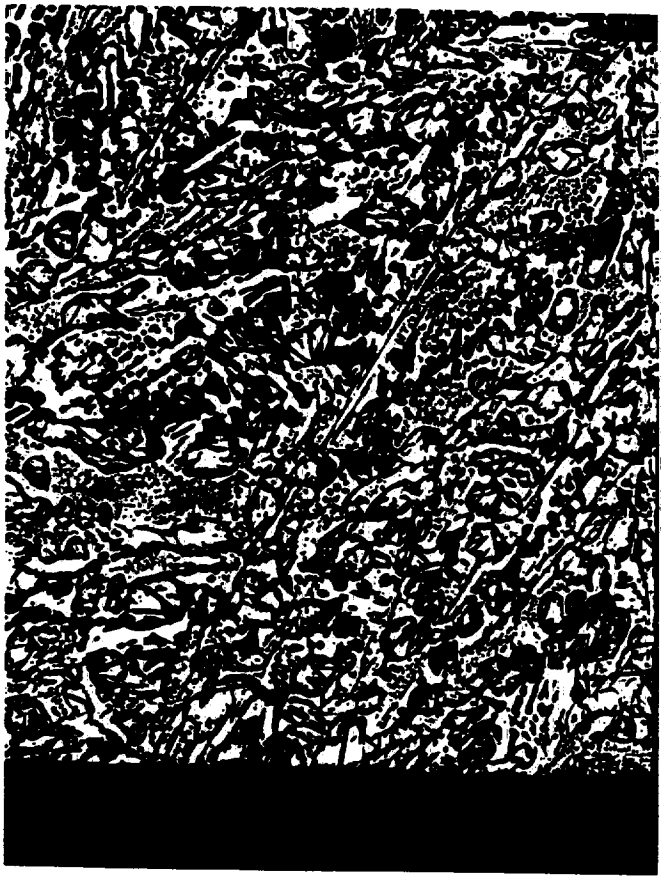


Photo 3373

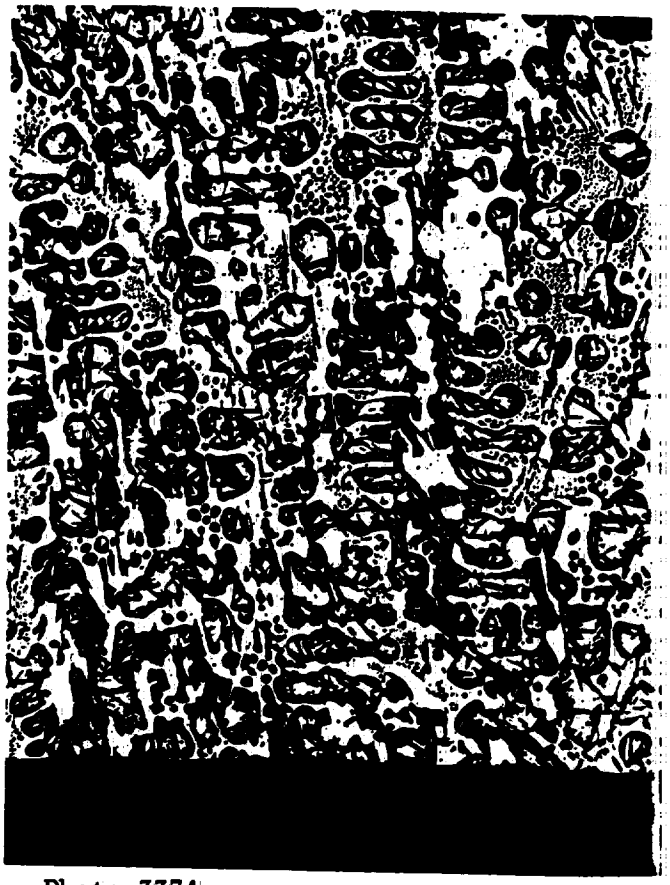


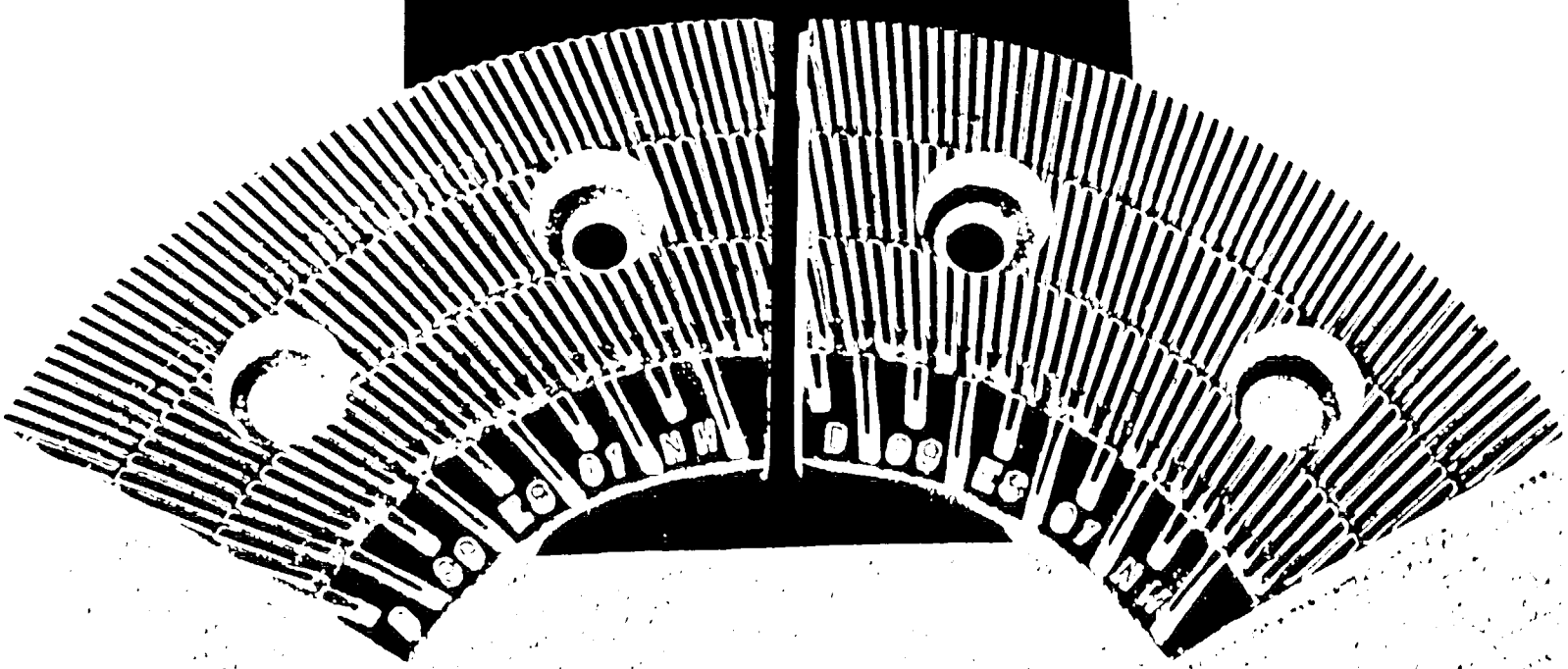
Photo 3374

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PROJECT NO. W 4

PHOTO NO. 3372



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